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AM25M009

Advanced Fluid Mechanics

Assignment - 3

①

① Stokes's first problem for stopping the plate from uniform motion.

the boundary conditions will be

$$u(0, t) = 0$$

$$u(\infty, t) = 0$$

$$u(y, 0) = \bar{u} \rightarrow \text{Before stopping.}$$

Continuing from the class notes.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad \text{and} \quad \frac{\partial^2 u}{\partial t^2} = \nu \frac{\partial^2 u}{\partial y^2}$$

$$\text{let } \eta = \frac{y}{2\sqrt{\nu t}}$$

$$F(\eta) = \frac{u}{\bar{u}}$$

$$u = F(\eta) \bar{u} \quad \text{--- (1)}$$

$$\frac{\partial u}{\partial t} = F'(\eta) \bar{u} \frac{\partial \eta}{\partial t} = F'(\eta) \bar{u} \frac{1}{2\sqrt{\nu}} t^{-1/2}$$

$$\left| \frac{\partial \eta}{\partial t} = \frac{-\frac{1}{2} t^{-3/2}}{t^{-1/2}} = -\frac{1}{2} t^{-1} \right|$$

$$\frac{\partial u}{\partial t} = \frac{1}{2} F'(\eta) \bar{u} \frac{1}{\sqrt{\nu t}} \quad \left| \frac{\partial u}{\partial t} = \frac{F'(\eta) \bar{u} \eta}{2t} \right| \quad \text{--- (1)}$$

$$\frac{\partial u}{\partial y} = \bar{u} F'(\eta) \frac{1}{2\sqrt{\nu t}} \quad \left| \frac{\partial^2 u}{\partial y^2} = \bar{u} F''(\eta) \frac{1}{4\nu t} \right| \quad \text{--- (2)}$$

$$(1) = \nu \times (2) \quad \Rightarrow \quad \frac{\partial u}{\partial t} - \frac{F'(\eta) \bar{u} \eta}{2t} = \bar{u} \frac{F''(\eta)}{4\nu t}$$

$$-2[F'(\eta)\eta] = F''(\eta)$$

$$[F''(\eta) + 2F'(\eta)\eta] = 0 \quad \text{--- (3)}$$

$$\frac{\partial F'(\eta)}{\partial \eta} + 2\eta F'(\eta) = 0$$

$$\frac{\partial F'(\eta)}{F'} = -2\eta d\eta \quad \Rightarrow \quad \ln F'(\eta) = -\eta^2 + A$$

$$\Rightarrow F'(\eta) = e^{-\eta^2 + A} = F'(\eta) = \underbrace{e^{-\eta^2}}_{\text{constant}} e^A = F'(\eta) = A e^{-\eta^2}$$

$$F(\eta) = A e^{-\eta^2} d\eta \quad \Rightarrow \quad \boxed{F(\eta) = A \int_0^\eta e^{-\eta^2} d\eta + B}$$

Applying boundary conditions to find A, B

$$\text{I} - u(0, t) = 0 \quad \left| \begin{array}{l} y=0 \\ t=t \end{array} \right| \quad \eta=0 \quad \left| \begin{array}{l} F(\eta)=0 \\ \eta=\frac{y}{2\sqrt{t}} \\ F(\eta)=\frac{y}{2\sqrt{t}} \end{array} \right.$$

$$\text{II} - u(\infty, t) = \bar{u} \quad \left| \begin{array}{l} y=\infty \\ t=t \end{array} \right| \quad \eta=\infty \quad \left| \begin{array}{l} F(\eta)=1 \end{array} \right.$$

$$\text{III} - u(y, 0) = \bar{u} \quad \left| \begin{array}{l} y=y \\ t=0 \end{array} \right| \quad \eta=\infty \quad \left| \begin{array}{l} F(\eta)=1 \end{array} \right.$$

$$\text{from I} \quad 0 = A \int_0^0 e^{-\eta^2} d\eta + B \quad \Rightarrow \quad \boxed{B=0}$$

$$\text{from II} \quad 1 = A \int_0^\infty e^{-\eta^2} d\eta$$

$$A \frac{\sqrt{\pi}}{2} = 1$$

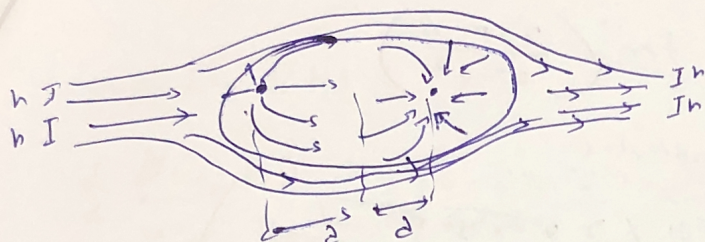
$$\boxed{A = \frac{2}{\sqrt{\pi}}}$$

$$\boxed{\begin{array}{l} \int_0^\infty e^{-x^2} = \frac{\sqrt{\pi}}{2} \\ \int_0^\infty e^{-x^2} = \frac{\sqrt{\pi}}{2} \end{array}}$$

$$\text{so it becomes} \quad F(\eta) = \frac{2}{\sqrt{\pi}} \int_0^\eta e^{-\eta^2} d\eta$$

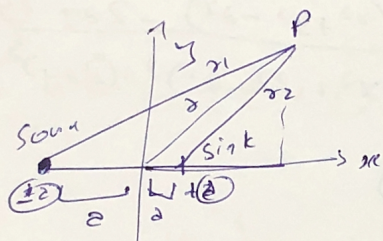
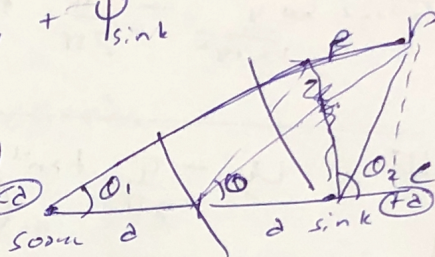
$$\text{so } u = \bar{u} F(\eta) \quad \boxed{u = \bar{u} \left(\frac{2}{\sqrt{\pi}} \int_0^\eta e^{-\eta^2} d\eta \right)}$$

② Rankine oval



$$\psi = \psi_{\text{uniform flow}} + \psi_{\text{source}} + \psi_{\text{sink}}$$

$$U \sin \theta + \frac{Q}{2\pi} \theta - \frac{Q}{2\pi} \theta_{\text{sink}}$$



$$\tan \theta_1 = \frac{y}{x+a}$$

$$\text{or } \tan \theta_1$$

$$\tan \theta_1 = \frac{y \sin \theta_1}{x \cos \theta_1 - a}$$

$$\psi = m \theta_2 - m \theta_1$$

$$\tan \theta_2 = \frac{y \sin \theta_2}{x \cos \theta_2 + a}$$

$$-\frac{\psi}{m} = \theta_1 - \theta_2 = \theta$$

$$\tan\left(-\frac{\psi}{m}\right) = \frac{\frac{y \sin \theta}{x \cos \theta - a} - \frac{y \sin \theta}{x \cos \theta + a}}{1 + \frac{y^2 \sin^2 \theta}{x^2 \cos^2 \theta - a^2}}$$

total ψ

$$\psi_{\text{total}} = U \sin \theta - \frac{Q}{2\pi} \left(\tan^{-1} \left(\frac{2ay \sin \theta}{x^2 - a^2} \right) + \tan^{-1} \left(-\frac{\psi}{m} \right) \right) = \frac{2ay \sin \theta}{x^2 - a^2}$$

$$Q = 2\pi \gamma V x$$

for stagnation streamline
 $x = b$ $\theta = \pi$

$$\psi = \frac{Q}{2\pi \gamma}$$

$$\psi_{\text{stag}} = \frac{Q}{2\pi \gamma}$$

$$-\frac{\psi}{m} = \tan^{-1} \left(\frac{2ay \sin \theta}{x^2 - a^2} \right)$$

$$\psi = -m \tan^{-1} \left(\frac{2ay \sin \theta}{x^2 - a^2} \right)$$

$$\psi = -\frac{Q}{2\pi \gamma} \tan^{-1} \left(\frac{2ay \sin \theta}{x^2 - a^2} \right)$$

finding half length
 $0 = 0 \quad 0 \leq \pi$

~~$b = \frac{Q}{2\pi U}$~~

$$\psi_{\text{total}} = U_0 \sin \theta - \frac{Q}{2\pi} \tan^{-1} \left(\frac{2a \sin \theta}{r^2 - a^2} \right)$$

Converting to cartesian coordinates.

$$U \sqrt{x^2 + y^2} \frac{y}{\sqrt{x^2 + y^2}} - \frac{Q}{2\pi} \tan^{-1} \left(\frac{2a \sqrt{x^2 + y^2} \frac{y}{\sqrt{x^2 + y^2}}}{x^2 + y^2 - a^2} \right)$$

$$\boxed{\psi = U_y - \frac{Q}{2\pi} \tan^{-1} \left(\frac{2ay}{x^2 + y^2 - a^2} \right)}$$

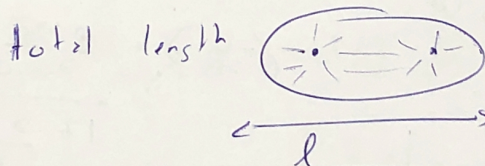
$$\frac{\partial \psi}{\partial y} = U - \frac{Q}{2\pi} \frac{2a(x^2 + y^2 - a^2) - 2ay \cdot 2y}{(x^2 + y^2 - a^2)^2 + (2ay)^2}$$

at $y=0$

$$u = U - \frac{Q}{2\pi} \frac{2a(x^2 - a^2)}{(x^2 - a^2)^2}$$

$u=0$ (stagnation points)

$$U = \frac{Q}{2\pi} \frac{2a(x^2 - a^2)}{(x^2 - a^2)^2} \quad \left| \quad x = \pm \sqrt{a^2 + \frac{Qa}{\pi U}} \right|$$



$$= 2 \sqrt{a^2 + \frac{Qa}{\pi U}}$$

for width, $x=0, y=h \quad \psi=0$

$$0 = Uh - \frac{Q}{2\pi} \tan^{-1} \left(\frac{2ah}{h^2 - a^2} \right)$$

$$Uh = \frac{Q}{2\pi} \tan^{-1} \left(\frac{2ah}{h^2 - a^2} \right) \quad \left[\begin{array}{l} \text{By solving this we get } h \\ \text{by} \end{array} \right]$$

(5)

$$\nabla \cdot (\mathbf{v} \cdot \nabla) \mathbf{v} = v_j \frac{\partial v_i}{\partial x_j} = \frac{1}{2} \nabla (\mathbf{v} \cdot \mathbf{v}) - \mathbf{v} \times (\nabla \times \mathbf{v})$$

expanding RHS

$$\frac{1}{2} \frac{\partial (\mathbf{v} \cdot \mathbf{v})}{\partial x_j} = \frac{1}{2} \left[v_k \frac{\partial v_k}{\partial x_j} + v_k \frac{\partial v_{1k}}{\partial x_j} \right] = \frac{v_k \partial v_k}{\partial x_j}$$

$$\underbrace{\nabla \times (\nabla \times \mathbf{v})}_{\text{vector}} = \epsilon_{jmn} v_m (\nabla \times \mathbf{v})_n$$

$$\underbrace{\quad}_{\text{vector}} = \epsilon_{lmn} v_m \epsilon_{nab} \frac{\partial v_b}{\partial x_a}$$

$$= \epsilon_{lmn} \epsilon_{nab} \epsilon_{nab} v_m \frac{\partial v_b}{\partial x_a} = \begin{pmatrix} \delta_{1a} \delta_{2b} - \delta_{1b} \delta_{2a} \\ \delta_{2a} \delta_{3b} - \delta_{2b} \delta_{3a} \\ \delta_{3a} \delta_{1b} - \delta_{3b} \delta_{1a} \end{pmatrix} v_m \frac{\partial v_b}{\partial x_a}$$

$$= v_b \frac{\partial v_b}{\partial x} - v_b \frac{\partial v_b}{\partial x} v_k \frac{\partial v_k}{\partial x} = v_m \frac{\partial v_m}{\partial x_1} - v_m \frac{\partial v_k}{\partial x_m}$$

totally.

$$v_k \frac{\partial v_k}{\partial x_j} - v_m \frac{\partial v_m}{\partial x_1} + v_m \frac{\partial v_k}{\partial x_m}$$

changing $k \rightarrow m$
 $j \rightarrow 1$

$$= v_k \frac{\partial v_j}{\partial x_k} = \boxed{(\mathbf{v} \cdot \nabla) \mathbf{v}}$$